Real-Time Video Inference on 8× NVIDIA RTX PRO 5000

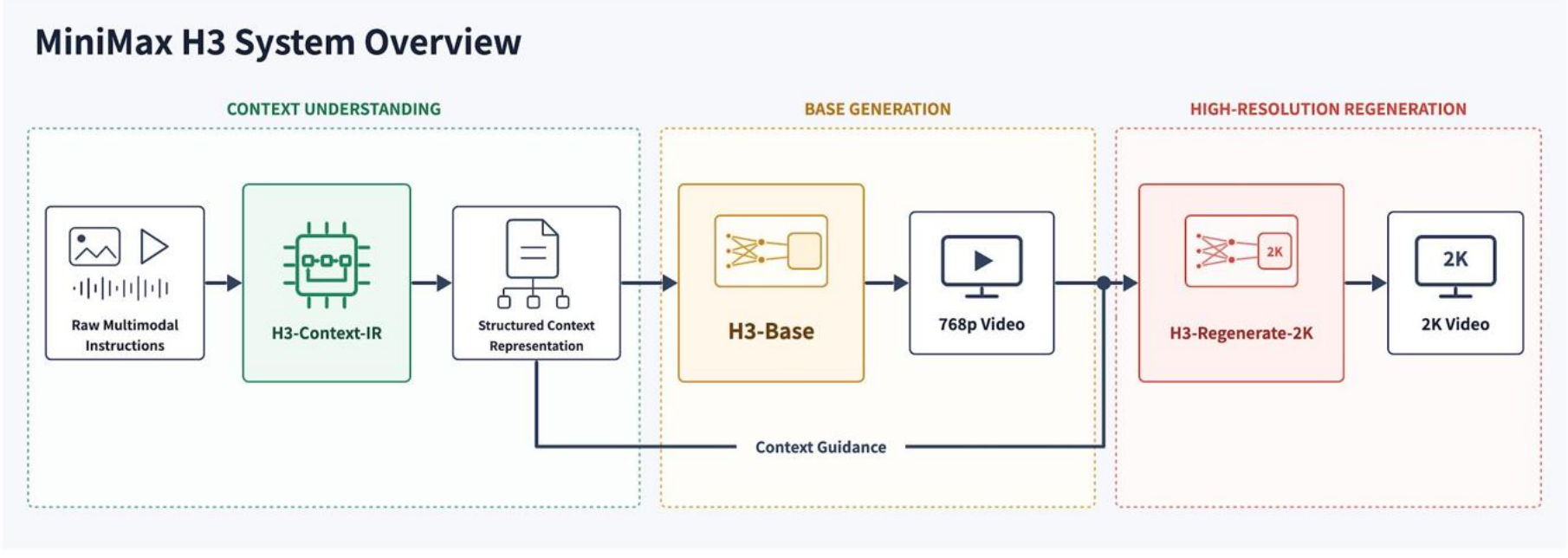
MiniMax H3 with vLLM-Omni

Shunyang Li | Public Technical Brief | October 2026



Real-Time Video Generation Is a Distributed-Systems Problem

For each request, end-to-end latency must stay within the generated clip's playback window while the evaluated 60K–1M-token workload reshapes memory, kernels, communication volume, and graph reuse.



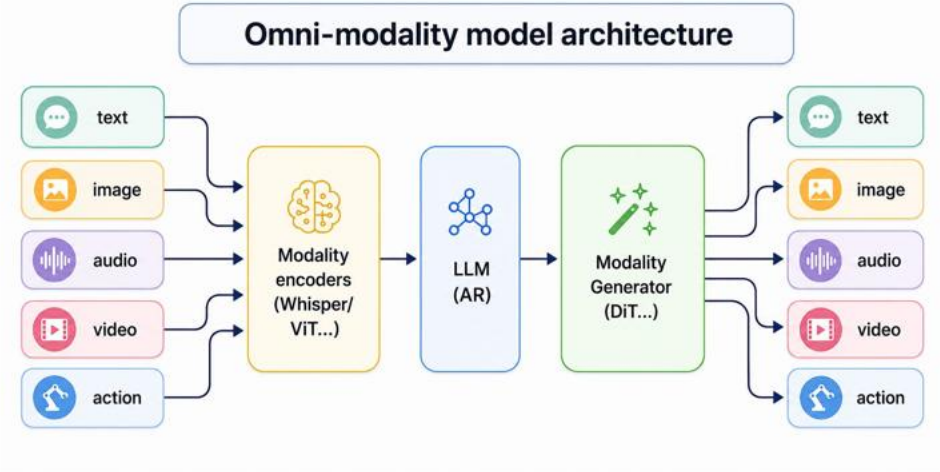
EVALUATED PATH · 8 DIT STEPS × 50 BLOCKS = 400 REPEATED BLOCK EXECUTIONS

Across SP8, the evaluated path repeatedly couples compute with Q/K/V and Gate/O exchange; launch gaps, transport fences, rank imbalance, and the decoder tail accumulate on the critical path, while the optimized system must preserve the reference output contract.

Sources: vLLM-Omni v0.26 public deck, slide 23 · author evaluated workload envelope and eight-GPU execution path

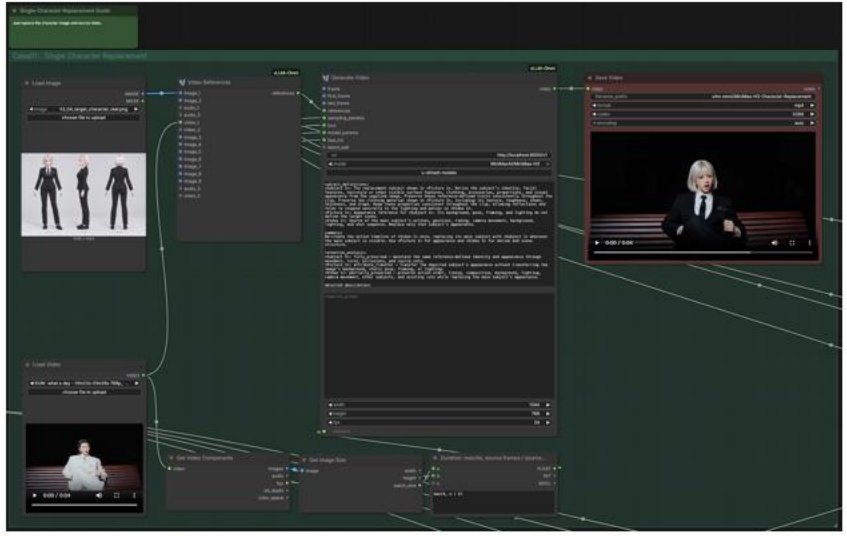
vLLM-Omni Connects Visual Workflows to Multi-GPU Serving

ComfyUI remains the authoring surface; vLLM-Omni owns serving, stage placement, batching, and multi-GPU execution



OFFICIAL vLLM-OMNI ARCHITECTURE

One serving layer spans omni-modal inputs, autoregressive reasoning, and DiT generation.



COMFYUI WORKFLOW · PR #9194

MiniMax-H3 character replacement calls the vLLM-Omni Generate Video endpoint; PR #9194 remains open.

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)

Frontier Models · The Cards

One Blackwell architecture (CC 12.0/12.1) spans several very different deployment envelopes

SKU	CC / SMs	Memory	Power	Interconnect
RTX PRO 5000 Blackwell 8-GPU baseline	12.0 / 110	72 GB GDDR7 ECC 48 GB variant	**350 W configured*** **300 W public TBP**	PCIe Gen5 ×16 no GPU NVLink
RTX PRO 6000 Blackwell WS / Server	12.0 / 188	96 GB GDDR7 ECC	600 W	PCIe Gen5 ×16 no GPU NVLink
GeForce RTX 5090	12.0 / 170	32 GB GDDR7	575 W	PCIe Gen5 ×16 NVLink: no
DGX Spark · GB10	12.1 / 48	128 GB unified LPDDR5X	140 W SoC 240 W system	NVLink-C2C in SoC 200 Gb/s ConnectX-7

SHARED BY DISCRETE CARDS · BLACKWELL FP4 · GDDR7 · PCIe GEN5 · NO GPU-TO-GPU NVLINK

Watch sustained clocks, not peak TOPS: controlled tensor-heavy tests reported SwPowerCap in 94.8–100% of valid samples; one rank stayed at 1.942 GHz and ran the same fine-attention input 6.78% slower. A 600 W-class design offers more headroom, but still requires a matched chassis A/B.

*350 W is the local configured/enforced limit; NVIDIA publicly specifies 300 W TBP for RTX PRO 5000.

Sources: NVIDIA product specifications · CUDA GPU Compute Capability · local system inventory and controlled clock study